

CANDIDATE ASSESSMENT SUBMISSION	Role: AI Automation Intern (PPO Track)
Firm: Meridian Advisory (Finance & Compliance)	Company: Suvitt Service Solutions LLP (LLPIN: ACG-0284)
Candidate Name: Om S Habib (Applied via Internshala)	Recipient: Aditi Singh, HR (aditi.singh@suvitt.com)
Subject Line: AI Automation Intern - Assessment - Om S Habib via Internshala	Evaluation Date: Wednesday, 19 August 2026

Executive Summary & Overview

This submission presents an end-to-end technical, operational, and architectural design for **Meridian Advisory**, a compliance and finance advisory firm handling statutory tax filings, document reviews, and regulatory deadlines. The design bridges two core operational systems: an enterprise ticketing platform and incoming client email communications, while embedding strict deterministic guardrails to eliminate AI hallucination risks surrounding dates, money, and regulatory commitments.

Part A - AI Prompting Skill

A1. Production System Prompt Design

Meridian requires an AI assistant to process unstructured client emails and return a deterministic, validated JSON classification. The prompt below incorporates: (1) strict Pydantic JSON schema enforcement, (2) unambiguous taxonomy boundaries, (3) explicit uncertainty signaling via calibrated confidence scores (<0.60) and uncertainty reasons, (4) verbatim quote anchoring, and (5) negative constraints against guessing statutory dates.

```
You are the Senior Intake & Triage AI for Meridian Advisory, a finance and compliance advisory firm.
Your role is to read an incoming client email and return a strictly validated, structured JSON classification.
Accuracy is mission-critical: mistakes regarding dates, financial figures, or legal commitments carry severe legal penalties
.

### STRICT OPERATIONAL RULES:
1. GROUNDING & ANTI-HALLUCINATION:
  - Extract information ONLY from the verbatim text in the provided email.
  - NEVER guess, assume, or extrapolate dates, ticket numbers, monetary amounts, or client obligations.
  - If a statutory deadline or ticket ID is not explicitly written in the email, you MUST set the corresponding field to null.
  - NEVER look up general knowledge (e.g., standard GST or TDS deadlines) to fill in dates.

2. CLASSIFICATION TAXONOMY (Choose exactly ONE):
  - "NEW_SERVICE_REQUEST": Client is requesting a new advisory service, tax filing, or consultation.
  - "EXISTING_TICKET_UPDATE": Client is following up, replying, or inquiring about an ongoing engagement or existing ticket number.
  - "DOCUMENT_SUBMISSION": Client is providing files, statements, invoices, or records purely for the firm's records without requesting new work.
  - "GENERAL_INQUIRY": Client is asking a general informational question not tied to a specific filing or ticket.
  - "IRRELEVANT_OR_SPAM": Marketing, sales solicitations, newsletters, automated receipts, out-of-office replies, or forwarded non-business content.
  - "AMBIGUOUS_OR_INSUFFICIENT": The email lacks critical context, contains vague directives (e.g., "handle asap" without details), or references attachments that are absent.

3. URGENCY LEVELS: "CRITICAL" | "HIGH" | "MEDIUM" | "LOW" | "UNKNOWN"
  - CRITICAL: Imminent regulatory deadlines (<24 hours), legal notices, penalty threats, or severe client distress.
  - HIGH: Near-term deadlines (2-5 days) or explicit urgent follow-ups.
  - MEDIUM: Standard turnaround or routine active matter follow-up.
  - LOW: Routine records submission or non-urgent queries.

4. CONFIDENCE & UNCERTAINTY HANDLING:
  - Provide a numerical confidence_score between 0.00 and 1.00.
  - If vague, short (<10 words without context), or missing attachments, set is_confident: false, score < 0.60, and list uncertainty_reasons.

5. DATE EXTRACTION:
  - extracted_due_date must ONLY be captured if client explicitly mentions a target date or deadline in text.
  - Include exact verbatim quote in quote_text.
  - If relative (e.g., "end of month"), normalize to YYYY-MM-DD based on email timestamp, set is_relative: true.

### OUTPUT JSON SCHEMA:
{
  "classification_type": "NEW_SERVICE_REQUEST" | "EXISTING_TICKET_UPDATE" | "DOCUMENT_SUBMISSION" | "GENERAL_INQUIRY" | "IRRELEVANT_OR_SPAM" | "AMBIGUOUS_OR_INSUFFICIENT",
  "ticket_reference": {"has_ticket_id": boolean, "ticket_id": string | null, "quote_text": string | null},
  "urgency": "CRITICAL" | "HIGH" | "MEDIUM" | "LOW" | "UNKNOWN",
  "urgency_rationale": string,
  "extracted_due_date": {"has_date": boolean, "raw_text": string | null, "normalized_date": string | null, "is_relative": boolean, "is_statutory": boolean, "quote_text": string | null},
  "context_completeness": "COMPLETE" | "PARTIAL" | "INSUFFICIENT",
  "missing_elements": [string],
  "confidence_score": number,
  "is_confident": boolean,
  "uncertainty_reasons": [string],
  "recommended_action": "ROUTE_TO_TEAM_QUEUE" | "APPEND_TO_EXISTING_TICKET" | "ARCHIVE_DOCUMENT_RECORD" | "HUMAN_MANUAL_TRIAGE" | "AUTO_DISCARD"
}
```

A2. Test Your Own Prompt Against Sample Emails

Execution results for the three test emails (evaluated under reference timestamp: 2026-08-19 09:00:00 IST):

Email 1: "Hi team, please find attached our Q2 bank statement for your records. Thanks."

Structured Model Output:

- classification_type:** "DOCUMENT_SUBMISSION" | **urgency:** "LOW" | **confidence_score:** 0.96 (is_confident: true)
- extracted_due_date:** null (has_date: false) | **ticket_reference:** null
- recommended_action:** "ARCHIVE_DOCUMENT_RECORD"
- Downstream Verification:** Deterministic MIME parser verifies file attachment exists before archiving.

Email 2: "Following up on ticket #4521 - any update on the GST filing? We need this closed by end of month."

Structured Model Output:

- classification_type:** "EXISTING_TICKET_UPDATE" | **ticket_id:** "#4521" (quote: "ticket #4521")
- urgency:** "HIGH" (active statutory filing follow-up) | **confidence_score:** 0.94 (is_confident: true)
- extracted_due_date:** raw_text: "end of month", normalized_date: "2026-08-31", is_relative: true, is_statutory: false
- recommended_action:** "APPEND_TO_EXISTING_TICKET"
- Downstream Verification:** Flagged as client target date, not statutory GST deadline (August 20).

Email 3: "URGENT!! Please handle asap." (No other content, and no attachment.)

Structured Model Output:

- **classification_type:** "AMBIGUOUS_OR_INSUFFICIENT" | **urgency:** "HIGH" (stress keywords detected)
- **confidence_score:** 0.20 (is_confident: false) | **missing_elements:** ["Subject matter", "Task instruction", "Attachment"]
- **uncertainty_reasons:** ["Zero substantive information", "No attachment detected", "Vague directive"]
- **recommended_action:** "HUMAN_MANUAL_TRIAGE"
- **Downstream Verification:** Triggers outbound call alert to Senior Account Lead; generates zero automated ticket commitments.

A3. Production Failure Analysis & Automated Detection Architecture

Specific Failure Mode: Historical Quoted-Thread Date Bleed & Relative Temporal Drift

Clients frequently reply to long email chains containing historical correspondence, outdated filings, or holiday footers (e.g., 'Office closed 15 August for Independence Day' or 'TDS paid on 07 July'). A common LLM failure is mistakenly extracting an outdated historical date or disclaimer as the active target due date for the current ticket.

Four-Layer Automated Detection Architecture (Zero Manual Reading Required):

1. **Pre-LLM Quoted-Text Stripping:** Deterministic RFC 3676 parser strips historical quote headers ('> ', 'On [date] wrote:') and signature footers before model ingestion.
2. **Verbatim Substring Assertion Tripwire:** Programmatic check asserts extracted_due_date.quote_text in unquoted_email_body. Fails automatically if not verbatim in new text.
3. **Temporal Sanity Bounds Filter:** Any extracted date where normalized_date < email_received_date or normalized_date > email_received_date + 90 days triggers an automatic anomaly exception.
4. **Ticketing CRM Entity Cross-Reference:** When a Ticket ID (#4521) is extracted, an API query verifies it exists and belongs to the sender's client domain, routing mismatches to human triage.

Part B - Workflow Design

B1. End-to-End Workflow Architecture

The workflow enforces a strict division of responsibility: deterministic rules handle security, thread continuity, and regulatory lookups, while AI is strictly constrained to natural language comprehension.

Workflow Stage	Mechanism	Engine Type	Design Rationale
1. Thread Matching	Inspect In-Reply-To, References headers and regex ticket ID.	Deterministic	100% precision. Prevents duplicate tickets on existing email chains.
2. Client Auth	Match sender email domain against Meridian CRM directory.	Deterministic	Verifies authorized client identity, active engagement contracts, and lead.
3. Intent & Urgency	Execute System Prompt (A1) on unquoted message body.	AI-Based (LLM)	Natural language understanding: separates filings, records, inquiries, and spam.
4. Safety Tripwires	JSON Schema validation, confidence threshold (>= 0.80), text check.	Deterministic	Catches hallucinations, format deviations, and ambiguous inputs instantly.
5. Statutory Calendar	Map identified service type to master regulatory compliance calendar.	Deterministic	Absolute Rule: Statutory dates come from official schedules, never AI.
6. Human Approval Gate	Team lead sign-off on scope, dates, and external commitments.	Human Gate	Prevents unverified commitments or inaccurate filings from reaching client.

B2. Explain Your Reasoning (Human-Approval Boundary)

Constraint: 150 words or fewer (Exact word count: 138 words)

We placed the human-approval boundary strictly before publishing external client commitments, activating statutory filing dates, and finalizing ticket closures.

Getting this boundary wrong in either direction carries severe consequences:

- **Shifting the line toward full automation (Under-supervision):** An AI misinterpreting an email could enter an incorrect statutory filing deadline or silently close an unresolved request. The firm risks missed regulatory deadlines, crippling statutory tax penalties, legal malpractice liability, and irreparable client churn.
- **Shifting the line toward manual review of everything (Over-supervision):** Forcing team leads to manually approve routine thread updates, spam filtering, and document archiving drowns senior advisors in administrative trivia. This causes employee burnout, SLA breaches on genuine client emergencies, and defeats the productivity purpose of automation.

The chosen boundary automates operational plumbing while reserving professional human judgement for regulatory and legal commitments.

B3. Scale It (Four Teams Architecture)

Constraint: 100 words or fewer (Exact word count: 94 words)

We would build **one parameterized workflow** governed by a configuration-driven architecture.

The core plumbing: email ingestion, thread matching, attachment validation, SLA tracking, error trapping, and exception routing: is universal across Tax, Legal, Operations, and Client Servicing. Rather than duplicating pipelines, team-specific logic is decoupled into external configuration tables (defining inbox webhooks, routing queues, statutory calendar lookups, and approval hierarchies).

Building four separate workflows creates four times the maintenance overhead, technical debt, and risk of logic drift. A single parameterized engine ensures centralized security, unified observability, and rapid firm-wide feature deployment.

Part C - Reporting and Data Logic

Evaluation Reference Date: Wednesday, 19 August 2026 | Working Week: Standard 5-day week (Monday to Friday), zero holidays.

C1. Data Quality Issues Audit (All Deliberate Traps Identified)

#	Issue Category	Ticket IDs	Defect Description & Risk Analysis
1	Exact Duplicates	T-101, T-105	Identical fields across client, status, created (10-Aug), next action (13-Aug), and statutory (25-Aug). Double-counts work and wastes staff hours.
2	Inconsistent Naming	T-101, T-102, T-105, T-110	Same client under 3 names: 'Acme Textiles', 'acme textile', 'ACME Textiles Pvt Ltd'. Breaks CRM aggregation and client SLA reporting.
3	Missing Action Date	T-102, T-107	Open tickets with NULL Next Action Date. Stalled without operational tracking. (Also blank on closed T-103, T-108).
4	Missing Statutory Date	T-106, T-103	Compliance tickets lacking regulatory deadlines; creates major blind spots for statutory compliance monitoring.
5	Missing Owner Column	All (T-101 to T-110)	The mandated 'Assigned Owner' column is 100% missing from the dataset. Results in zero individual accountability.
6	Overdue Statutory Dates	T-104, T-107	Statutory deadlines already expired on open tickets: T-104 (due 01-Aug, 18 days overdue!), T-107 (due 18-Aug, 1 day overdue). Severe penalty exposure.
7	Weekend Dates	T-104, T-109, T-110	Dates on Saturdays/Sundays: T-104 created/due Sat 01-Aug; T-109 action Sat 15-Aug; T-110 created Sun 09-Aug. Violates 5-day firm calendar.
8	Chronological Contradiction	T-104	Next Action Date (03-Aug) set 2 days AFTER Statutory Due Date (01-Aug). Logically impossible workflow sequence.
9	Ambiguous Status	T-106	Status set to 'Pending' with next action 14-Aug (3 working days overdue). Unclear if paused or client-dependent.

C2. Working-Day SLA Breach Calculations

Firm Rule: 'Tickets not moved forward within two working days are considered overdue.' Standard 5-day week (Sat-Sun excluded).

1. Calculation for Ticket T-102 (Client: 'acme textile'):

- Status: Open | Created: Tue 11-Aug-2026 | Next Action Date: [BLANK] | Statutory Due: Thu 20-Aug-2026 (Tomorrow!)
- Working days elapsed: Wed 12-Aug (Day 1), Thu 13-Aug (Day 2: SLA limit), Fri 14-Aug (Day 3: Breach), Mon 17-Aug (Day 4), Tue 18-Aug (Day 5), Wed 19-Aug (Day 6: Today).
- Result: 6 working days open without action date (4 working days overdue). CRITICAL SLA BREACH** (filing due tomorrow!).

2. Calculation for Ticket T-107 (Client: 'Bluewave Foods'):

- Status: Open | Created: Thu 30-Jul-2026 | Next Action Date: [BLANK] | Statutory Due: Tue 18-Aug-2026 (Yesterday: Overdue!)
- Working days elapsed: 31-Jul (1 day), 3-7 Aug (5 days), 10-14 Aug (5 days), 17-19 Aug (3 days) = **14 working days**.
- Result: 14 working days open without action date (12 working days overdue). MASSIVE SLA BREACH.**

3. Operational SLA Overdue Calculations (Lapsed Next Action Dates):

- T-101 & T-105:** Next Action 13-Aug (Thu). Lapsed working days to 19-Aug: Fri 14, Mon 17, Tue 18, Wed 19 = **4 working days overdue. SLA BREACH.**
- T-104:** Next Action 03-Aug (Mon). Lapsed working days to 19-Aug = **12 working days overdue. CATASTROPHIC BREACH.**
- T-110:** Next Action 11-Aug (Tue). Lapsed working days to 19-Aug: 12, 13, 14, 17, 18, 19 Aug = **6 working days overdue. SLA BREACH.**
- T-106 (Pending):** Next Action 14-Aug (Fri). Lapsed working days: Mon 17, Tue 18, Wed 19 = **3 working days overdue. SLA BREACH.**

C3. One-Page Daily Exception Report (Top Three Lines)

- [REGULATORY DEADLINE BREACH]:** 2 Open Tickets Have Passed Statutory Due Date: T-104 (Crest Pharma: 18 days overdue) and T-107 (Bluewave Foods: 1 day overdue). Immediate partner intervention required to mitigate penalty and legal exposure.
- [IMMINENT FILING AT RISK]:** 1 Open Ticket Due Tomorrow Has No Assigned Next Action: T-102 (acme textile: Statutory Due Date 20-Aug-2026). 6 working days stalled; immediate emergency assignment required before 17:00 IST.
- [INTERNAL SLA & DATA INTEGRITY]:** 5 Active Tickets in Severe SLA Breach (>2 working days stalled: T-101, T-105, T-110, T-106, T-107); 1 Confirmed Duplicate Pair (T-101 / T-105). Action: Merge duplicates and enforce next action updates today.

Part D - AI Judgement and Risk

(Constraint: Answer each question in three to five sentences.)

D1. Statutory Deadline Error: Accountability & Preventive System Design

Accountability is shared between the system designer and the team lead: the designer failed to implement basic data validation and grounding guardrails, while the team lead breached supervisory fiduciary duty by treating unverified, probabilistic AI text as an authoritative source of truth. Architecturally, an AI should never be permitted to unilaterally extract or update statutory compliance deadlines from conversational email text without verification. Instead, the workflow must deterministically cross-reference client filing scopes against an authoritative, master regulatory tax calendar maintained by the firm's compliance officers. Furthermore, any AI-suggested modification to a statutory date must mandate a dual-control human-in-the-loop sign-off with verbatim source citations before writing to the system of record. Finally, the automated reporting engine should have featured a disparity tripwire flagging whenever an extracted date diverged from the regulatory master calendar.

(Exact sentence count: 5 sentences)

D2. Deliberate Non-AI Automation Task & Technical Justification

A concrete task where I would deliberately refuse to use AI is calculating statutory payroll tax withholdings, late filing penalty interest, and compound tax liabilities. While a modern large language model can parse tax tables and perform arithmetic, generative models are non-deterministic, probabilistic token predictors that are fundamentally prone to rounding errors, hallucination, and numerical drift. Statutory financial computations require 100% mathematical precision, instantaneous execution speed, and an airtight, reproducible audit trail for tax authorities. Implementing this calculation through a deterministic script written in Python or SQL guarantees zero calculation error, total legal auditability, and zero API token cost.

(Exact sentence count: 4 sentences)

D3. Universal Inbox Access Governance & Scoping

Before granting inbox access, I would ask what specific operational bottleneck necessitates full mailbox scanning, what data privacy and zero-retention agreements are established with the AI provider, and how privileged communications (such as attorney-client privilege, whistleblower reports, and internal HR matters) are sequestered. I would also ask what legal liabilities arise under data protection regulations (such as GDPR or India's DPDP Act) if unredacted client financial records are exposed to model pipelines. Rather than granting broad personal inbox access, the minimum access I would request is restricted read-only webhook access scoped exclusively to designated shared departmental inboxes (such as tax-filings@meridian.com or compliance-support@meridian.com). Finally, this integration must run through an on-premise PII redaction layer that strips banking credentials, passwords, and sensitive identifiers before payload delivery to the LLM.

(Exact sentence count: 4 sentences)

Part E - Communication

E1. Handover Note to Non-Technical Team Lead (168 words, <= 200 words)

Subject: How your new morning compliance report works (and what to do if it ever fails)

Hi Sarah, to ensure you never have to guess whether your automated morning report ran, we have established a simple **"Heartbeat Rule"**: Every business morning by **8:30 AM**, you will receive an email titled **"Meridian Daily Compliance Digest - [Date]"**. If that email has not arrived in your inbox by **8:35 AM**, or if you receive a notification titled **"ALERT: Daily Report Generation Incomplete"**, the automated run has encountered an issue.

If that happens, please take two quick steps: (1) **Open Your Saved Backup View:** In your ticketing dashboard, click the bookmark titled **"00_Daily_Exceptions_Backup"**. This live view displays the exact same overdue tickets and imminent statutory deadlines, updating independently of the email service. (2) **Alert the Automation Team:** Send a quick message to our internal Teams channel **#ops-automation** or text me directly: **"Daily report missing for [Date]"**.

Our team will diagnose the pipeline immediately and deliver a manual PDF digest within 15 minutes. Your client operations can continue smoothly without interruption! - AI Automation Team

E2. Client-Facing Role Explanation (Single Cohesive Paragraph)

While tools like ChatGPT are conversational assistants that generate text when prompted, an AI Automation role at Meridian Advisory designs the secure, invisible operational infrastructure that guarantees your financial filings and compliance obligations are executed flawlessly. Rather than simply typing questions into a chatbot, this role builds interconnected systems that automatically route incoming client correspondence, match deadlines against official statutory tax schedules, detect missing documentation before it causes filing delays, and track team response times against strict quality standards. Crucially, this architecture is engineered with enterprise safeguards: AI is used to eliminate manual administrative friction, but every regulatory commitment, monetary figure, and tax filing is governed by deterministic rules and signed off by qualified human advisors. In short, ChatGPT is an ad-hoc drafting tool, whereas AI automation is an institutional-grade reliability engine designed to protect your firm from missed deadlines, ensure strict confidentiality, and deliver predictable compliance outcomes.

Submission Checklist & AI Usage Disclosure

- [X] **Part A:** Prompt, test outputs, and failure analysis | [X] **Part B:** Workflow stages, reasoning (138 words), scaling (94 words)
- [X] **Part C:** Data issues table, working-day SLA math, top 3 lines | [X] **Part D:** Three judgement answers (3-5 sentences each)
- [X] **Part E:** Handover note (168 words) & client explanation (1 paragraph) | [X] **Disclosure:** Detailed below
- Disclosure of AI Tool Usage:** *In adherence to candidate instructions: 'You may use AI to complete this assessment. If you do, tell us where and how you used it. This is not a penalty: how you use AI is part of what we are evaluating.'*
- **AI Tools & Runtimes:** Gemini 3.8 Flash via Google Antigravity Agentic IDE; local Python 3.14 environment for mathematical date verification scripts.
 - **Application:** (1) Part A: drafted JSON schemas with strict grounding constraints. (2) Part B: structured workflow stages table. (3) Part C: Python script verified all working-day math; AI synthesized findings into audit table. (4) Parts D & E: iteratively refined drafts to satisfy strict sentence and word counts while maintaining professional executive tone.